

CHEME 5660: Nomenclature and Worked Explanations

Fall 2026 | Main lectures through Week 6 | Updated September 29, 2026

This reference follows the lectures from L1b through L6b. Each section connects its symbols with a calculation: why we use the equation, how the steps fit together, and what the result tells us. The notation tables collect meanings and units for lookup.

Questions from the engagement surveys

The larger themes in your feedback appear in the lecture sections where we develop them:

- **L3a - Volatility clustering:** What persists when direction is hard to predict? We use the lecture's SPY/AMD figure to distinguish a turbulent period from an isolated large movement.
- **L3b - Forecasting versus pricing:** Why does one tree use two probabilities? We use the same two future prices to compute a forecast and the price of a replicated payoff.
- **L4b - Mean growth versus drift:** Which average does each parameter describe? We explain the half-variance correction and compare expected and median prices.
- **L4b - Model purpose and limits:** What can a price model help us decide? After developing trade NPV and GBM, we calculate a target probability, change an input, and explain how to check the forecast.
- **L5b - Correlated shocks:** How do independent inputs produce correlated movements? We multiply a two-asset loading matrix by a normal vector, identify the shared input, and check the covariance and time units.
- **Throughout - Equations and notation:** How do I read the symbols? Use the lecture tables alongside the calculations and the final context index when a letter is reused.

Each of the five conceptual explanations includes a short check with an answer. The numerical examples are illustrative; the figure is reproduced from L3a. Example-specific variables and optional advanced derivations remain defined in their own notebooks.

The notebook requires no code execution. Its companion PDF has the same content. We will extend the reference as the course progresses; please flag an unclear symbol or step with its lecture label.

Reading the Notation

In the tables, **1** means dimensionless and **yr** means year. Matrix dimensions appear in the definition column. Read the subscript and lecture context along with the letter.

Symbol	Meaning and context	Units
$\ln x, \log x$	Natural logarithm; price ratios inside a logarithm are dimensionless.	1
$\mathbb{E}[X]$	Expected value of a random quantity X .	Same as X
$\text{Var}(X)$	Variance; the square of the standard deviation.	Units of X squared
$\text{Cov}(X, Y)$	Covariance of two random quantities.	Units of X times units of Y
$\mathbb{P}, \mathbb{Q}$	Real-world and risk-neutral probability laws (L3b).	Probabilities: 1
$\hat{x}$	Usually an estimate from observations. The augmented design matrix $\hat{\mathbf{X}}$ is a separate convention (L4b, L6b).	Same as x
g'	Sample mean growth rate; the prime marks a mean in these lectures.	yr^{-1}
$\mathbf{x}, \mathbf{A}$	Bold symbols denote vectors or matrices; subscripts identify entries.	Context dependent
$\mathbf{A}^\top, \mathbf{A}^{-1}$	Transpose and inverse, respectively. An inverse requires a square, nonsingular matrix.	Context dependent
$\mathbf{1}, \mathbf{I}_M$	Vector of ones of the required length; $M \times M$ identity matrix.	1
$\ \mathbf{x}\ _2^2$	Sum of squared entries of $\mathbf{x}$.	Entry units squared

Read an expression as a calculation. For example, $g_j = \ln(S_j/S_{j-1})/\Delta t$ starts with a price ratio, takes its natural logarithm, and divides by the elapsed time. The price units cancel in the ratio; dividing by years gives a growth rate in inverse years. The L3a section works through this calculation.

Separate inputs from outputs. In a forecast we supply an initial price, parameter estimates, and a holding period. The model then supplies a distribution of future prices. An expectation averages over that distribution; a probability adds up the portion satisfying a stated event, such as exceeding a sale-price target.

Distinguish an estimate from the quantity it describes. The sample mean g' estimates the population mean growth rate μ_g . A hat usually marks a fitted quantity, such as $\hat{\mu}_g$.

L1a is the orientation slide deck. Its option-preview symbol δ denotes option-price sensitivity to the underlying price; its detailed development comes later in the course.

L1b: Time Value of Money and Net Present Value

We move cash flows to a common date before comparing them. The accumulation factor moves money forward in time, and its inverse discounts a future payment to time zero.

Symbol	Meaning and context	Units
P, F_j	Present value at time zero and future value after j compounding periods.	USD
y	Nominal annual yield, quoted as a decimal, with its compounding convention.	yr^{-1}
n	Compounding frequency: number of periods per year.	yr^{-1}
j, ℓ, N	Period indices and final cash-flow index; j/n is elapsed time in years.	Counts
T	Elapsed time or investment horizon; nT is the number of periods.	yr
i_ℓ	Effective rate during compounding period ℓ .	1
$\mathcal{D}_{j,0}$	Accumulation factor from date 0 to period j ; for a constant yield, $(1 + y/n)^j$.	1
$\mathcal{D}_{j,0}^{-1}$	Discount factor from period j to date 0.	1
$g(t), g$	Time-varying or constant continuously compounded growth rate.	yr^{-1}
g_y	Continuous growth rate equivalent to the quoted yield: $n \log(1 + y/n)$.	yr^{-1}
$\mathcal{A}$	Abstract asset represented by its dated net cash flows.	Cash-flow collection
$c_{j,k}, \mathbf{c}_j$	Nonnegative amount of cash flow k at date j ; vector of the m_j amounts.	USD
$\nu_{j,k}, \boldsymbol{\nu}_j$	Cash-flow direction: +1 for money received, −1 for money paid.	1
m_j	Number of cash-flow components at date j .	Count
$\bar{c}_j, \bar{\mathbf{c}}$	Signed net cash flow at date j ; vector of net cash flows at dates $0, \dots, N$.	USD
$\mathcal{D}_{\star,0}^{-1}$	Vector of the $N + 1$ discount factors for those cash-flow dates.	1
NPV_0	Net present value: sum of discounted signed cash flows.	Time-0 USD

How do the cash-flow symbols become a value? At date j , multiply each cash amount $c_{j,k}$ by its direction $\nu_{j,k}$ and add the results. Then discount that date's net cash flow to time zero:

$$\bar{c}_j = \sum_{k=1}^{m_j} c_{j,k} \nu_{j,k}, \quad \text{NPV}_0 = \sum_{j=0}^N \mathcal{D}_{j,0}^{-1} \bar{c}_j.$$

The first sum combines payments made on the same date. The second combines their present values across dates. For example, paying 100 USD today and receiving 110 USD in one year at a 5% annual benchmark gives $\text{NPV}_0 = -100 + 110/1.05 \approx 4.76$ USD. The positive result means that the payment exceeds the benchmark requirement by 4.76 USD in today's money. The discount factor is dimensionless, so each term in the NPV retains units of USD.

L2a: Treasury Bills, Notes, and Bonds

The yield convention determines the accumulation and discount factors. A bill's bank-discount quote uses a 360-day basis; the investment horizon T uses the year-fraction convention selected for valuation.

Symbol	Meaning and context	Units
V_B	Purchase price or modeled value of the Treasury security.	USD
V_P	Par (face) value paid at maturity.	USD
d	Annual bank-discount quote for a Treasury bill, entered as a decimal.	Annual quote, 360-day basis
D	Actual number of days remaining to bill maturity.	Days
T	Time to maturity under the selected valuation day-count convention.	yr
y, n, g_y	Nominal annual yield, compounding frequency, and equivalent continuous rate from L1b.	yr^{-1}
g_B	Realized continuously compounded growth for a bill held to maturity.	yr^{-1}
r_B	Bill's holding-period log return: $r_B = Tg_B$.	1
c	Nominal annual coupon rate as a decimal.	yr^{-1}
C	Cash coupon per payment date: $C = (c/n)V_P$.	USD
N	Number of remaining coupon payments; $N = nT$ on the lecture's coupon-date grid.	Count
$\mathcal{D}_{N,0}^{-1}$	Discount factor for the par repayment date.	1

From a bill quote to investment growth. The bank-discount quote expresses the discount as a fraction of par. We first calculate the purchase price, then use the purchase-to-par ratio to calculate growth over the holding period:

$$V_B = V_P \left(1 - d \frac{D}{360} \right), \quad g_B = \frac{1}{T} \ln \left(\frac{V_P}{V_B} \right).$$

For $V_P = 100$ USD, $d = 0.04$, and $D = 90$ days, the bill costs 99 USD. If we select $T = 90/365$ years for valuation, $g_B \approx 0.04076 \text{ yr}^{-1}$. The quote uses par and a 360-day basis; the growth calculation uses the amount invested and the selected holding-time convention.

What changes for a coupon bond? We receive coupons along the way as well as par at maturity. On a coupon date, with a constant discount yield, its value is the sum of those discounted payments:

$$V_B = \underbrace{V_P \mathcal{D}_{N,0}^{-1}}_{\text{present value of par}} + \underbrace{C \sum_{j=1}^N \mathcal{D}_{j,0}^{-1}}_{\text{present value of coupons}}.$$

Each discount factor applies to the date of its payment. This is the L1b cash-flow calculation with the bond's payment schedule supplied.

L2b: Yield Sensitivity, Duration, Convexity, and the Curve

Duration and convexity summarize a price change when the quoted yield changes. The spot curve supplies a separate discount factor for each payment date.

Symbol	Meaning and context	Units
$V_B(y, T)$	Security price as a function of yield and time to maturity.	USD
X_j	Promised cash flow at payment j , including par at maturity.	USD
$t_j = j/n$	Time of payment j .	yr
$V_{B,y}$	Yield-sensitivity magnitude: $-\partial V_B/\partial y$.	USD · yr
$V_{B,T}$	Bill maturity-sensitivity quantity: $-\partial V_B/\partial T$.	USD/yr
$V_{B,c}$	Coupon-rate sensitivity: $\partial V_B/\partial c$.	USD · yr
D_{mod}	Modified duration: $-(1/V_B)(dV_B/dy)$.	yr
D_{mac}	Macauley duration: payment times averaged using their shares of present value.	yr
$K(y)$	Convexity: $(1/V_B)(d^2V_B/dy^2)$.	yr ²
$\Delta y, \Delta V_B$	Finite changes in quoted yield and price. Enter yield changes as decimals; 1 basis point is 0.0001.	yr ⁻¹ ; USD
$\bar{y}$	Yield to maturity: one constant yield reproducing the market price from promised payments.	yr ⁻¹
$D(0, t_j)$	Spot-curve discount factor for a payment at t_j . This is the date-indexed counterpart of $\mathcal{D}_{j,0}^{-1}$.	1
c_{par}	Coupon rate that prices a hypothetical bond at par under the spot curve.	yr ⁻¹

How much will a yield change move the price? Modified duration gives the first-order price sensitivity; convexity corrects for curvature. With the lecture's nominal-yield convention:

$$D_{\text{mod}} = \frac{D_{\text{mac}}}{1 + y/n}, \quad \frac{\Delta V_B}{V_B} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}K(\Delta y)^2.$$

For $D_{\text{mod}} = 5$ yr, $K = 30$ yr², and a yield increase $\Delta y = 0.01$ yr⁻¹, the estimated price change is $-0.05 + 0.0015 = -0.0485$, or a 4.85% decrease. This local approximation is most useful for small yield changes.

How do we value payments with a yield curve? The curve supplies a discount factor $D(0, t_j)$ for each payment date. Adding the discounted payments gives the price; setting that price equal to par and solving for the coupon rate gives:

$$V_B = \sum_{j=1}^N X_j D(0, t_j), \quad c_{\text{par}} = n \frac{1 - D(0, t_N)}{\sum_{j=1}^N D(0, t_j)}.$$

The par coupon makes the present value of the coupons equal the difference between par and the present value of the final par repayment.

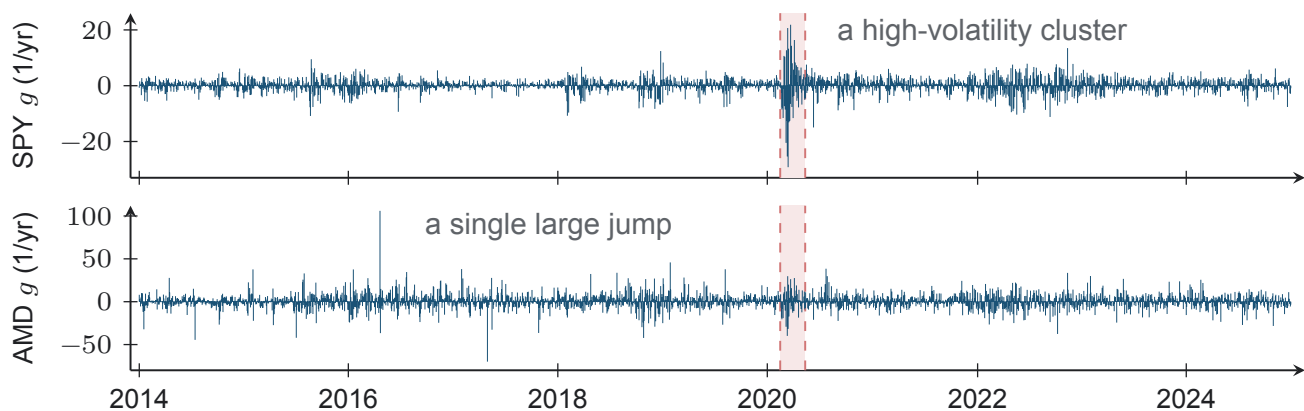
L3a: Observed Growth Rates and Stylized Facts

What persists when direction is hard to predict? Volatility clustering describes persistence in the **size** of movements. A turbulent period can contain both gains and losses. We measure direction with signed growth rates and movement size with their absolute values.

First convert consecutive prices into growth rates. For asset i , the observation interval Δt_j divides the log price ratio:

$$g_{j,j-1}^{(i)} = \frac{1}{\Delta t_j} \ln \left(\frac{S_j^{(i)}}{S_{j-1}^{(i)}} \right).$$

A change from 100 to 101 USD/share over $1/252$ year gives a log return $\ln(1.01) \approx 0.00995$ and a growth rate $252 \ln(1.01) \approx 2.508 \text{ yr}^{-1}$. The rate is large because we divide a small one-day change by a small fraction of a year. If that same log-price change occurred on each of 252 trading days, the accumulated log change would be 2.508. Here, we use the per-year scale to record one observed day; a forecast requires a model for the days that follow.



What do we see? This is the figure from L3a's *Volatility Clustering* section. The shaded March-2020 period contains many large movements of both signs in SPY and AMD. The isolated April-2016 AMD spike is one large jump. Clustering concerns a sequence of large movements, so we distinguish that turbulent period from the isolated spike.

The lecture measures persistence using **autocorrelation**, the linear association between observations separated by a specified number of time steps. Applying it to signed growth rates asks whether positive or negative deviations tend to recur. Applying it to growth magnitudes asks whether large or small movements tend to recur. Positive magnitude autocorrelation across several lags is evidence of persistence in movement size.

Why does this matter for a model? A fixed volatility uses the same uncertainty scale after calm and turbulent days. Clustering suggests that recent movement sizes contain information about upcoming risk. When we reach the lattice and GBM models, we can ask whether their assumptions preserve or discard that information.

Check. If signed growth has little autocorrelation but magnitude has persistent positive autocorrelation, can we infer the sign of tomorrow's change? **Answer:** this evidence supports persistence in movement size. It supplies no directional prediction or fixed date when a turbulent episode must end.

L3a: Notation for the Worked Explanation

Symbol	Meaning and context	Units
$S_j^{(i)}$	Observed share price of asset i at observation j .	USD/share
Δt_j	Time from price observation $j - 1$ to j ; Δt denotes equal spacing.	yr
$g_{j,j-1}^{(i)}, g_t$	Observed log-price change divided by the observation interval.	yr^{-1}
r_t	Log return over one interval: $r_t = \Delta t g_t$.	1
$\bar{g}$	Excess growth relative to the benchmark: $g - g_y$.	yr^{-1}
T	Number of price observations in this lecture; there are $T - 1$ growth observations.	Count
g', r'	Sample means of growth rates and log returns.	$\text{yr}^{-1}; 1$
σ_g	Sample standard deviation of interval growth rates.	yr^{-1}
$\sigma_r(\Delta t)$	Standard deviation of interval log returns: $\Delta t \sigma_g$.	1
σ_{ann}	Annualized log-return volatility: $\sqrt{\Delta t} \sigma_g$, under square-root-of-time scaling. Under GBM this is the diffusion volatility σ used in L4b.	$\text{yr}^{-1/2}$
$X = g , x$	Growth magnitude and a tail threshold in the same units.	yr^{-1}
$\alpha, \hat{\alpha}_k$	Power-law tail index and its Hill estimate using the largest k magnitudes.	1
$X_{(i)}, n, k$	Magnitudes ranked largest first, total tail-analysis sample size, and number of upper-order observations used.	Growth units; counts
$L(x)$	Slowly varying factor in the tail expression $\mathbb{P}(X > x) \sim L(x)x^{-\alpha}$; evaluated in a fixed unit convention.	Tail-scaling factor
ν	Degrees of freedom of a Student- t distribution used to discuss heavy tails.	Count parameter
$\hat{R}(k)$	Sample growth-rate autocovariance at lag k .	yr^{-2}
$\rho(k)$	Growth-rate autocorrelation: $\hat{R}(k)/\hat{R}(0)$.	1
$\overline{ g }_i, \hat{\rho}_i(k)$	Mean growth magnitude for asset i and its magnitude autocorrelation at lag k .	$\text{yr}^{-1}; 1$
k (autocorrelation)	Separation between observations; elapsed separation is $k\Delta t$.	Lag steps

Reading the two autocorrelations. For signed growth, $\hat{R}(k)$ averages products of deviations from the sample mean, and $\rho(k)$ divides by the zero-lag variance. For magnitude autocorrelation $\hat{\rho}_i(k)$, apply the same operation to $|g_t^{(i)}|$ after subtracting its own mean. The normalization cancels the growth units, allowing us to compare persistence on the same scale.

Growth variability versus diffusion volatility. Multiplying growth rates by Δt gives interval log returns, so $\sigma_r(\Delta t) = \Delta t \sigma_g$. Under GBM, the log-return standard deviation is also $\sigma\sqrt{\Delta t}$. Equating these gives $\sigma = \sqrt{\Delta t} \sigma_g$. As an illustrative daily-scale conversion, $\sigma_g \approx 3.1749 \text{ yr}^{-1}$ corresponds to $\sigma \approx 0.20 \text{ yr}^{-1/2}$. Scaling observed quantities at one interval is a change of units; extrapolating variability across horizons adds model assumptions.

L3b: Binomial Price Lattices

Why does one tree use two probabilities? Suppose a share costs 100 USD today and will be worth either 120 or 90 USD after one step. The up and down factors are $u = 1.20$ and $d = 0.90$. Keep these prices fixed while we ask two different questions.

Forecasting: how often might each outcome occur? Let the estimated real-world up probability be $p = 0.65$. Weighting the two prices by their forecast probabilities gives:

$$\mathbb{E}_{\mathbb{P}}[S_1] = 0.65(120) + 0.35(90) = 109.50 \text{ USD/share.}$$

This average describes the forecast distribution. Its quality depends on whether the estimated probabilities describe later outcomes adequately.

Pricing: what should a specified payoff cost today? Suppose a claim pays $H_u = 20$ USD after an up move and $H_d = 0$ after a down move. Let the risk-free accumulation factor over the same step be $R_f = 1.02$. With frictionless trading, short positions allowed, and borrowing or lending at this rate, no-arbitrage requires the risk-neutral weight q to satisfy:

$$qu + (1 - q)d = R_f \implies q = \frac{R_f - d}{u - d} = \frac{1.02 - 0.90}{1.20 - 0.90} = 0.40.$$

The condition $d < R_f < u$ gives $0 < q < 1$. We use these weights to average the claim's payoffs and then discount one step:

$$V_0 = \frac{qH_u + (1 - q)H_d}{R_f} = \frac{0.40(20) + 0.60(0)}{1.02} \approx 7.84 \text{ USD.}$$

Why is that the price? We can build the claim's payoff from the share and a loan. The claim's payoff differs by $20 - 0 = 20$ USD between the two states, while the share price differs by $120 - 90 = 30$ USD/share. Holding $20/30 = 2/3$ of a share therefore reproduces the payoff difference. In the down state those shares are worth 60 USD and the claim pays zero, so we need a loan repayment of 60 USD. Borrowing $60/1.02$ USD today supplies that repayment after one step. After one step the shares are worth 80 or 60 USD, and the loan repayment is 60 USD. The net payoff is therefore 20 or zero, exactly the claim's payoff. The initial cost is $100(2/3) - 60/1.02 \approx 7.84$ USD. A different claim price would let us trade the claim against this replicating position for an arbitrage profit under the stated assumptions.

Thus p describes forecast frequencies, while q values a payoff through a trading argument. Risk-neutral pricing does not assert that the stock is risk-free or that 40% of actual moves will be upward.

Check. Change only p from 0.65 to 0.50. What changes? **Answer:** the forecast expected price becomes 105 USD/share. The replicated payoff still costs about 7.84 USD because the two possible share prices and the borrowing rate are unchanged.

L3b: Notation for the Worked Explanation

Symbol	Meaning and context	Units
$S_0, S_{t,k}$	Initial price and node price after t steps with k up moves.	USD/share
t, k, N	Lattice level, number of up moves, and total number of simulated steps.	Counts
u, d	Up and down multiplicative price factors.	1
p	Up-move probability under the real-world law $\mathbb{P}$.	1
q	Up-move probability under the risk-neutral law $\mathbb{Q}$.	1
Δt	Length of one lattice step.	yr
R_f	Risk-free gross growth factor for one step: $e^{g_f \Delta t}$.	1
H_u, H_d, V_0	Claim payoffs in the two next-step states and its present price.	USD
U, D	Collections of observed up and down price multipliers used in calibration.	Multipliers: 1
X_j	Bernoulli indicator: 1 for an up move at step j , 0 for a down move.	1
g_j, μ_g, σ_g^2	One-step growth rate, its model mean, and its model variance.	yr ⁻¹ ; yr ⁻² for variance
M (growth bound)	Bound on growth magnitude: $\max\{ \ln d , \ln u \}/\Delta t$.	yr ⁻¹
$\mathcal{F}_j$	Information from the first j moves.	Information set
$\tau, \rho_g(\tau)$	Lag in lattice steps and growth autocorrelation at that lag.	Count; 1

Extending the forecast tree. With fixed multipliers and independent moves, a path containing k up moves in t steps ends at $S_0 u^k d^{t-k}$. There are $\binom{t}{k}$ orders of those moves, so the node's price and total probability are:

$$S_{t,k} = S_0 u^k d^{t-k}, \quad \mathbb{P}(k \text{ up moves}) = \binom{t}{k} p^k (1-p)^{t-k}.$$

For two steps and $p = 0.65$, the middle node is $100(1.20)(0.90) = 108$ USD/share. It can be reached by up-down or down-up, so its probability is $2(0.65)(0.35) = 0.455$. This explains why the binomial coefficient appears. The three node probabilities sum to one.

L4a: Quotes, Trade NPV, and Lattice Targets

We evaluate a long position at a scheduled sale time. Scaled NPV measures the discounted gain relative to the initial investment.

Symbol	Meaning and context	Units
a_t, b_t	Best ask and best bid at time t .	USD/share
s_t, m_t	Quoted spread $a_t - b_t$ and midpoint $(a_t + b_t)/2$.	USD/share
n_0	Number of shares bought at entry and held to sale.	Shares
S_0, S_T	Entry and terminal model share prices.	USD/share
$N, T = N\Delta t$	Number of lattice steps and elapsed holding time.	Count; yr
g_y	Continuous benchmark growth rate used to discount sale proceeds.	yr^{-1}
ρ_T	Scaled trade NPV: $\text{NPV}_0 / (n_0 S_0)$.	1
$\rho_\star$	Investor's target scaled NPV, with $\rho_\star > -1$.	1
$\rho(k)$	Scaled NPV at the terminal node with k up moves.	1
K_N	Random number of up moves in N steps; binomial with parameters N, p .	Count
$\tau(\rho_\star)$	Real-valued up-count threshold obtained by solving $\rho(k) > \rho_\star$.	Step-count threshold
$\kappa, k_{\min}$	$\kappa = \lfloor \tau \rfloor + 1$ is the first strictly successful integer; $k_{\min}$ clips it to $[0, N + 1]$.	Counts
m, f_j, p_j	Number of branches, positive price factor on branch j , and its probability in an m -ary lattice.	Count; 1; 1
$x_j, \mathbf{x}, \mathbf{X}_t$	Number of branch- j moves, count vector, and random count vector after t steps; $\sum_j x_j = t$.	Counts
$S_t(\mathbf{x})$	Price at count state $\mathbf{x}$: $S_0 \prod_{j=1}^m f_j^{x_j}$.	USD/share
L_t	Number of count states: $\binom{t+m-1}{m-1}$. Distinct count states can have the same price.	Count

Which lattice outcomes beat our target? Buying n_0 shares costs $n_0 S_0$ today; their sale brings $n_0 S_T$ at time T . Ignoring dividends and trading costs, discount and divide by the purchase cost:

$$\rho_T = \frac{-n_0 S_0 + n_0 S_T e^{-g_y T}}{n_0 S_0} = \frac{S_T}{S_0} e^{-g_y T} - 1.$$

At a terminal lattice node with k up moves, substitute $S_T/S_0 = u^k d^{N-k}$. Taking logs of $\rho(k) > \rho_\star$ and collecting the terms in k gives, for $u > d > 0$:

$$k > \tau(\rho_\star) = \frac{\ln(1 + \rho_\star) + g_y T - N \ln d}{\ln(u/d)}.$$

The up count is an integer, so the first strictly successful count is $\lfloor \tau \rfloor + 1$. Add the binomial probabilities for that count and all higher counts. If $\tau = 2$ exactly, success starts at three up moves; two only meets the target. Clipping the first count to $[0, N + 1]$ represents the cases where every node or no node succeeds. This turns a trading requirement into a probability over the model's terminal outcomes.

L4b: Single-Asset Geometric Brownian Motion

Which average do drift and mean growth describe? Geometric Brownian motion uses drift μ for expected price growth and volatility σ for the size of random changes. With constant parameters, the price equation is:

$$\frac{dS(t)}{S(t)} = \mu dt + \sigma dW(t).$$

Here $W(t)$ is a Wiener process; its increment over Δt has variance Δt . The random price increment has zero conditional mean, so averaging the price equation gives $d\mathbb{E}[S(t)]/dt = \mu\mathbb{E}[S(t)]$. Solving this mean equation gives $\mathbb{E}[S_T] = S_0 e^{\mu T}$. We will compare that expression with the average obtained from the log-price solution to find how drift relates to mean growth.

For the price at a fixed time T , the solution in terms of mean growth μ_g is:

$$\ln(S_T/S_0) = \mu_g T + \sigma\sqrt{T} Z, \quad Z \sim \mathcal{N}(0, 1).$$

Averaging this equation makes the random term vanish because $\mathbb{E}[Z] = 0$. Thus μ_g is expected log growth per unit time. To average the price itself, we exponentiate first. The normal identity $\mathbb{E}[e^{aZ}] = e^{a^2/2}$, valid for any real number a , gives:

$$\mathbb{E}[S_T] = S_0 e^{\mu_g T} \mathbb{E}[e^{\sigma\sqrt{T}Z}] = S_0 e^{(\mu_g + \sigma^2/2)T}.$$

Comparing this result with $\mathbb{E}[S_T] = S_0 e^{\mu T}$ gives $\mu = \mu_g + \sigma^2/2$, or $\mu_g = \mu - \sigma^2/2$. This is the **half-variance correction**. Averaging prices and averaging their logarithms are different operations; the normal exponential average supplies the precise difference for GBM. The L4b derivation obtains the same correction from Itô's lemma.

A numerical comparison. Let $S_0 = 100$ USD/share, $\mu = 0.08 \text{ yr}^{-1}$, $\sigma = 0.20 \text{ yr}^{-1/2}$, and $T = 1$ year. Then $\mu_g = 0.06 \text{ yr}^{-1}$, giving:

$$\begin{aligned} \mathbb{E}[S_T] &= 100e^{0.08} \approx 108.33 \text{ USD/share}, \\ \text{median}(S_T) &= 100e^{0.06} \approx 106.18 \text{ USD/share}. \end{aligned}$$

The median splits the possible prices into equal lower and upper halves. It follows from the median $Z = 0$ and the increasing exponential function. The right tail pulls the expected price above the median; neither summary identifies a particular future path.

Which number belongs in a simulation? The equivalent updates are:

$$S_{t+\Delta t} = S_t e^{\mu_g \Delta t + \sigma\sqrt{\Delta t} Z} = S_t e^{(\mu - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} Z}.$$

The same draw Z gives the same price in both forms. If we estimate μ_g from observed growth rates, the correction is already included; subtracting it again would lower the simulated mean growth incorrectly.

Check. Hold $\mu = 0.08 \text{ yr}^{-1}$ fixed and increase σ to $0.40 \text{ yr}^{-1/2}$. **Answer:** $\mu_g = 0$, the expected one-year price remains about 108.33 USD/share, and the median becomes 100 USD/share. Higher volatility separates the two averages further.

The L4b notation table, after the trade application, collects these symbols and the simulation-array notation.

L4b: What Can a Price Model Help Us Decide?

What can we decide if we cannot predict the exact price? Suppose we buy a share for $S_0 = 100$ USD/share and plan to sell after $T = 1$ year. We want the proceeds to exceed an investment growing at the benchmark rate $g_y = 0.02 \text{ yr}^{-1}$. We assume a scheduled sale, no dividends or trading costs, and a constant benchmark rate. Discounting the sale price S_T gives the scaled net present value:

$$\rho_T = \frac{-S_0 + S_T e^{-g_y T}}{S_0}.$$

The target $\rho_T > 0$ means that the discounted proceeds exceed the purchase cost. Solving this inequality tells us which future prices count as successes:

$$\rho_T > 0 \iff S_T > S_0 e^{g_y T} = 100 e^{0.02} \approx 102.02 \text{ USD/share}.$$

Where does the probability come from? Under geometric Brownian motion (GBM), the log price ratio at the sale date is normally distributed. Let $\mu_g = 0.06 \text{ yr}^{-1}$ be mean growth and $\sigma = 0.20 \text{ yr}^{-1/2}$ be volatility. The model writes:

$$\ln(S_T/S_0) = \mu_g T + \sigma \sqrt{T} Z, \quad Z \sim \mathcal{N}(0, 1).$$

Here Z is a standard normal random variable. Substitute the model into the target condition, then subtract the mean and divide by the positive scale $\sigma\sqrt{T}$:

$$\begin{aligned} \mu_g T + \sigma \sqrt{T} Z &> g_y T, \\ Z &> \frac{(g_y - \mu_g)T}{\sigma \sqrt{T}} = \frac{0.02 - 0.06}{0.20} = -0.20. \end{aligned}$$

The normal cumulative distribution function $\Phi(z)$ gives the probability to the left of z . Our target uses the area to the right, so:

$$\mathbb{P}(\rho_T > 0) = 1 - \Phi(-0.20) \approx 0.5793.$$

We obtain a **57.9% model probability of beating the benchmark**, and a 42.1% probability of finishing at or below it. This supplies a risk comparison for the specified trade. It concerns the sale date; exiting when a price first reaches a boundary is a different trading rule.

How much should we trust the answer? If mean growth were 0.02 yr^{-1} instead, the threshold for Z would be zero and the success probability would be 50%. Estimated inputs therefore matter. The model also assumes constant parameters and independent, normally distributed log-price increments. Clustering or heavier tails in the data can change the risk beyond what a new mean estimate captures.

To check the forecast, fit on earlier observations and save probabilities for later, fixed-horizon trades before their outcomes are known. Over many comparable cases, compare the forecast probabilities with the frequency of the same event. For a prediction interval at one date, we check whether the price at that date lies inside it. To check a band intended to contain an entire path, every observed price must remain inside its bounds; one crossing is enough to fail that check. The L5a out-of-sample example develops that distinction.

Check. Keep the parameters above but require a 5% scaled NPV. What changes? **Answer:** the price threshold becomes $100(1.05)e^{0.02} \approx 107.12$, and the probability falls to about 48.2%. We raise the target before asking the model for its probability.

L4b: Notation for the Worked Explanation

Symbol	Meaning and context	Units
$S(t), S_{t_j}$	Continuous-time share price and its value on the simulation grid.	USD/share
μ	GBM drift: $\mathbb{E}[S_T] = S_0 e^{\mu T}$.	yr ⁻¹
μ_g	Mean log growth rate: $\mu - \sigma^2/2$.	yr ⁻¹
σ	GBM diffusion volatility.	yr ^{-1/2}
$W(t), dW(t)$	Wiener process and its infinitesimal increment. An increment over Δt has variance Δt .	yr ^{1/2}
Z, Z_j	Standard normal shock; simulation uses independent shocks across time steps.	1
$\Delta t, t_j = j\Delta t, T$	Step length, elapsed grid time, and terminal horizon.	yr
N, M (simulation)	Number of steps and number of Monte Carlo paths.	Counts
S	Simulated price array with $N + 1$ rows and M paths as columns.	USD/share
g_j, g', σ_g	Observed interval growth, its sample mean, and sample standard deviation.	yr ⁻¹
$\hat{\mu}_g$	Estimate of mean log growth, commonly g' .	yr ⁻¹
$\hat{\sigma}$	Estimated GBM volatility: $\sigma_g \sqrt{\Delta t}$.	yr ^{-1/2}
$\hat{\mu}$	Estimated drift: $\hat{\mu}_g + \hat{\sigma}^2/2$.	yr ⁻¹
$\Phi(z)$	Standard normal cumulative distribution function.	Probability: 1
$z_\star$	Standardized threshold for the scheduled-sale scaled-NPV target.	1

A general scaled-NPV target. The preceding worked trade example uses $\rho_\star = 0$. For any target $\rho_\star > -1$, substitute the price solution into scaled NPV and take logs. With $T > 0$ and $\sigma > 0$:

$$\begin{aligned} \rho_T > \rho_\star &\iff (\mu_g - g_y)T + \sigma\sqrt{T}Z > \ln(1 + \rho_\star), \\ &\iff Z > z_\star = \frac{\ln(1 + \rho_\star) - (\mu_g - g_y)T}{\sigma\sqrt{T}}. \end{aligned}$$

We subtract the mean benchmark-adjusted log growth and divide by its standard deviation. The resulting threshold is dimensionless. The probability of exceeding it is $1 - \Phi(z_\star)$; raising the target raises the threshold and reduces the probability.

L4b: Regression and Parameter Uncertainty

The lecture also fits a straight line to log price against time. These symbols describe that regression; L6b reuses the regression structure with different response and predictor variables.

Symbol	Meaning and context	Units
$\mathbf{y}$	Vector of the $N + 1$ observed log prices. A fixed price unit is implicit when taking log price.	1
$\hat{\mathbf{X}}$	Augmented design matrix: $N + 1$ rows, columns of ones and times t_j .	Columns: 1; yr
$\boldsymbol{\theta}$	Parameter vector $(\ln S_0, \mu_g)^\top$.	Entries: 1; yr ⁻¹
$\hat{\boldsymbol{\theta}}$	Least-squares estimate of the intercept and slope.	Same as parameters
ϵ_j	Cumulative stochastic deviation of log price from the line at time t_j .	1
$\mathbf{r}$	Fitted residual vector: $\mathbf{y} - \hat{\mathbf{X}}\hat{\boldsymbol{\theta}}$.	1
$n = N + 1, p = 2$	Number of regression observations and number of fitted parameters.	Counts
$\nu = n - p$	Residual degrees of freedom.	Count
$\hat{\sigma}_\epsilon^2$	Residual variance estimate under the lecture's simplifying regression-error model.	1
$\text{SE}(\hat{\theta}_j)$	Standard error of fitted coefficient j .	Same as coefficient
α	Significance level; the stated confidence level is $1 - \alpha$.	1
$t_{1-\alpha/2, \nu}$	Student- t quantile used in a two-sided coefficient interval.	1

What are we fitting, and what does the interval describe? For observation j , the lecture fits log price as an intercept plus mean growth times elapsed time. Stacking all observations gives:

$$\ln S_j = \ln S_0 + \mu_g t_j + \epsilon_j \quad \Longleftrightarrow \quad \mathbf{y} = \hat{\mathbf{X}}\boldsymbol{\theta} + \boldsymbol{\epsilon}.$$

Each row of the design matrix is $[1, t_j]$. Least squares chooses the intercept and slope to minimize the squared residuals. For a fitted coefficient $\hat{\theta}_j$, the standard error estimates its uncertainty; a smaller value indicates greater precision. The lecture's two-sided confidence interval is:

$$\hat{\theta}_j \pm t_{1-\alpha/2, \nu} \text{SE}(\hat{\theta}_j).$$

This is an interval for a model coefficient. A prediction interval for a future price must also account for future random price changes. The classical coefficient interval assumes independent, constant-variance, normally distributed errors. GBM log-price errors accumulate through time and share earlier shocks, so this simplifying regression-error assumption limits the interval's interpretation. Fitting a line alone does not validate its uncertainty calculation.

L5a: Trade Targets and Updating GBM Estimates

We retain the GBM and trade-NPV notation from L4a-L4b. The new symbols describe a terminal price target and exponential moving estimates of growth-rate moments.

Symbol	Meaning and context	Units
K	Terminal price threshold corresponding to target scaled NPV: $S_0(1 + \rho_*)e^{g_y T}$.	USD/share
ρ_T, ρ_*	Realized scaled trade NPV and the investor's target.	1
s, k	Entry observation index and subsequent update index.	Counts
g_k	Latest growth observation: $\ln(S_k/S_{k-1})/\Delta t$.	yr ⁻¹
λ	EMA decay factor, with $0 < \lambda < 1$.	1
H_{half}	Number of observations over which an old weight halves; $\lambda = 2^{-1/H_{\text{half}}}$.	Observation count
δ_k	Innovation relative to the previous mean: $g_k - m_{k-1}$.	yr ⁻¹
m_k	Exponentially weighted mean growth rate after observation k .	yr ⁻¹
v_k	Exponentially weighted growth-rate variance after observation k .	yr ⁻²
$\hat{\mu}_{g,k}$	Updated GBM mean log growth: m_k .	yr ⁻¹
$\hat{\sigma}_k$	Updated GBM volatility: $\sqrt{v_k \Delta t}$.	yr ^{-1/2}
$\hat{\mu}_k$	Updated GBM drift: $\hat{\mu}_{g,k} + \hat{\sigma}_k^2/2$.	yr ⁻¹
$\hat{\mu}_{g,0}, \hat{\sigma}_0$	Initial training estimates used to initialize m_s and v_s .	yr ⁻¹ ; yr ^{-1/2}

What changes when another price arrives? The new growth observation g_k differs from the previous mean by the innovation $\delta_k = g_k - m_{k-1}$. We retain a fraction λ of the old information and give the new observation weight $1 - \lambda$:

$$m_k = \lambda m_{k-1} + (1 - \lambda)g_k = m_{k-1} + (1 - \lambda)\delta_k.$$

For a 21-observation half-life, $\lambda = 2^{-1/21} \approx 0.9675$, so the new observation receives about 3.25% of the mean's weight. Half-life describes estimation memory; T describes how far ahead we forecast.

Why is the variance update different? Variance is measured about the updated mean. Measuring the old observations about this new center adds the squared shift of the mean to the old variance. Combining their contribution with the new observation gives:

$$v_k = \lambda [v_{k-1} + (m_{k-1} - m_k)^2] + (1 - \lambda)(g_k - m_k)^2.$$

The mean update gives $m_k - m_{k-1} = (1 - \lambda)\delta_k$, and the new observation's deviation is $g_k - m_k = \lambda\delta_k$. Substituting these two deviations and collecting terms yields:

$$\begin{aligned} v_k &= \lambda v_{k-1} + \lambda(1 - \lambda)^2 \delta_k^2 + (1 - \lambda)\lambda^2 \delta_k^2 \\ &= \lambda [v_{k-1} + (1 - \lambda)\delta_k^2]. \end{aligned}$$

Initialize $m_s = \hat{\mu}_{g,0}$ and $v_s = \hat{\sigma}_0^2/\Delta t$ from the training data. The updated growth variance has units yr⁻²; multiplying by Δt and taking a square root gives GBM volatility $\hat{\sigma}_k = \sqrt{v_k \Delta t}$. Each new forecast uses the updated estimates. In the course example, those estimates remain fixed over that forecast's horizon, so updating them does not by itself model future volatility clusters within the horizon.

L5b: Multiple-Asset GBM and Covariance

How can independent inputs produce correlated asset movements? Give the assets a shared random input. Suppose their volatility parameters are 0.20 and $0.30 \text{ yr}^{-1/2}$, with correlation 0.50 . The required covariance rate and a loading matrix that produces it are:

$$\mathbf{C} = \begin{bmatrix} 0.04 & 0.03 \\ 0.03 & 0.09 \end{bmatrix} \text{yr}^{-1}, \quad \mathbf{A} = \begin{bmatrix} 0.20 & 0 \\ 0.15 & \sqrt{0.0675} \end{bmatrix} \text{yr}^{-1/2}.$$

The first diagonal loading is $\sqrt{0.04} = 0.20$. The shared loading is $0.03/0.20 = 0.15$; the remaining loading is $\sqrt{0.09 - 0.15^2} = \sqrt{0.0675} \approx 0.259808$. These choices make $\mathbf{A}\mathbf{A}^\top = \mathbf{C}$.

Let Z_1 and Z_2 be independent standard normal variables. Write $\mathbf{X} = \mathbf{A}\mathbf{Z}$ for the two noise contributions before time scaling. Expanding the matrix product gives:

$$X_1 = 0.20Z_1, \quad X_2 = 0.15Z_1 + \sqrt{0.0675} Z_2.$$

Both assets use Z_1 , which produces their covariance. Independence removes the cross term containing Z_1Z_2 , and each input has variance one, so:

$$\begin{aligned} \text{Var}(X_1) &= 0.20^2 = 0.04, \\ \text{Var}(X_2) &= 0.15^2 + 0.0675 = 0.09, \\ \text{Cov}(X_1, X_2) &= 0.20(0.15) = 0.03. \end{aligned}$$

These values have units yr^{-1} . Dividing the covariance by $0.20(0.30)$ recovers correlation 0.50 . One draw illustrates the multiplication: if $\mathbf{Z} = (1, -1)^\top$, then $\mathbf{X} \approx (0.20, -0.109808)^\top \text{yr}^{-1/2}$. Positive correlation allows individual draws with opposite signs; it describes the joint distribution across draws.

How do these numbers enter a price step? For asset i , add mean log growth and the time-scaled noise in the exponent:

$$S_i(t + \Delta t) = S_i(t) \exp \left[\mu_{g,i} \Delta t + \sqrt{\Delta t} X_i \right], \quad \mu_{g,i} = \mu_i - C_{ii}/2.$$

For $\Delta t = 1/252 \text{ yr}$, starting prices of 100 USD/share, and both mean growth rates 0.06 yr^{-1} , the draw above gives next prices about 101.29 and 99.33 USD/share. The same vector is shared across assets within the step; we draw a fresh vector at the next step.

Why use a loading matrix? For independent standard normal inputs, $\text{Cov}(\mathbf{A}\mathbf{Z}) = \mathbf{A}\mathbf{A}^\top$. Multiplying by $\mathbf{C}$ itself would produce $\mathbf{C}\mathbf{C}^\top$, with the wrong covariance and units.

Check. Replace the loading 0.15 by -0.15 . **Answer:** both variances stay the same because they use squared loadings. Covariance becomes -0.03 yr^{-1} and correlation becomes -0.50 .

L5b: Notation for the Worked Explanation

Symbol	Meaning and context	Units
$\mathcal{P}, M = \mathcal{P} $	Asset set and number of assets.	Set; count
i, j, ℓ	Asset indices i, j and independent noise-source index ℓ .	Indices
$\mu_i, \mu_{g,i}$	Asset drift and mean log growth; $\mu_{g,i} = \mu_i - C_{ii}/2$.	yr^{-1}
$\mathbf{A}$	$M \times M$ loading matrix multiplying independent Wiener shocks.	$\text{yr}^{-1/2}$
$\mathbf{C} = \mathbf{A}\mathbf{A}^\top$	$M \times M$ instantaneous covariance-rate matrix.	yr^{-1}
$\sigma_i = \sqrt{C_{ii}}$	Diffusion volatility of asset i .	$\text{yr}^{-1/2}$
$W_\ell(t)$	Independent Wiener process for noise source ℓ .	$\text{yr}^{1/2}$
$\mathbf{X} = \mathbf{A}\mathbf{Z}$	Noise contributions before time scaling in the worked example; length M .	$\text{yr}^{-1/2}$
$\mathbf{g}_k, \boldsymbol{\mu}_g$	Length- M vectors of interval growth rates and their population means.	yr^{-1}
$\boldsymbol{\Sigma}_g$	Population covariance of the interval growth-rate vector; $M \times M$.	yr^{-2}
$\mathbf{Z}_k$	Length- M vector of independent standard normal shocks at step k .	1
$N, \mathbf{G}$	Number of aligned growth observations and the $N \times M$ data matrix.	Count; yr^{-1}
$\mathbf{g}^{(i)}$	Length- N column of growth observations for asset i .	yr^{-1}
$g'_i, \mathbf{g}'$	Sample mean for asset i and length- M vector of asset means.	yr^{-1}
$\tilde{\mathbf{G}}$	Column-centered growth matrix: $\mathbf{G} - \mathbf{1}(\mathbf{g}')^\top$.	yr^{-1}
$\hat{\boldsymbol{\Sigma}}_g$	Sample covariance of interval growth rates; $M \times M$.	yr^{-2}
$\hat{\mathbf{C}}$	Estimated GBM covariance rate: $\Delta t \hat{\boldsymbol{\Sigma}}_g$.	yr^{-1}
ρ_{ij}	Correlation of assets i and j ; covariance divided by their standard deviations.	1

Which covariance belongs in the model? Divide the log-price step by Δt to obtain the growth-rate vector. Fixed means contribute no covariance, and $\text{Cov}(\mathbf{Z}_k) = \mathbf{I}_M$, giving:

$$\mathbf{g}_k = \boldsymbol{\mu}_g + \frac{\mathbf{A}\mathbf{Z}_k}{\sqrt{\Delta t}}, \quad \boldsymbol{\Sigma}_g = \text{Cov}(\mathbf{g}_k) = \frac{\mathbf{C}}{\Delta t}.$$

To estimate that growth covariance, center each data column on its own mean. The (i, j) entry of $\tilde{\mathbf{G}}^\top \tilde{\mathbf{G}}$ adds the products of aligned deviations for assets i and j . Divide by $N - 1$, then multiply by the observation interval to estimate the covariance rate:

$$\hat{\boldsymbol{\Sigma}}_g = \frac{\tilde{\mathbf{G}}^\top \tilde{\mathbf{G}}}{N - 1}, \quad \hat{\mathbf{C}} = \Delta t \hat{\boldsymbol{\Sigma}}_g.$$

Thus growth covariance has units yr^{-2} , covariance rate has units yr^{-1} , and the log-price shock covariance over a step is $\Delta t \mathbf{C}$, which is dimensionless. In the two-asset example, daily growth covariance is $252\mathbf{C}$ numerically; daily log-return covariance is $\mathbf{C}/252$ numerically. The input's definition determines the conversion.

L5b: Portfolio Weights and Dirichlet Sampling

Initial weights divide the budget among assets. With buy-and-hold trading, share counts stay fixed while the fractions of wealth change as prices move.

Symbol	Meaning and context	Units
W_0, W_t	Initial budget and portfolio wealth at time t .	USD
$w_i, \mathbf{w}$	Initial fraction in asset i and length- M allocation vector.	1
n_i	Shares purchased: $w_i W_0 / S_i(0)$.	Shares
$w_i(t)$	Current wealth fraction: $n_i S_i(t) / W_t$.	1
Δ^{M-1}	Simplex of long-only weights with $w_i \geq 0$ and $\sum_i w_i = 1$.	Set of allocations
α, α_i	Positive Dirichlet concentration parameters used to sample allocations.	1
α_0	Total concentration: $\sum_i \alpha_i$.	1
α (symmetric case)	Common concentration when every $\alpha_i = \alpha$.	1
$\mathbf{g}$	Length- M vector of asset growth rates over one interval.	yr ⁻¹
g_p	Linear one-period portfolio growth proxy: $\mathbf{w}^\top \mathbf{g}$.	yr ⁻¹
g'_p	Sample mean of the proxy: $\mathbf{w}^\top \mathbf{g}'$.	yr ⁻¹
$\sigma_{g,p}^2$	Proxy's sample variance: $\mathbf{w}^\top \hat{\Sigma}_g \mathbf{w}$.	yr ⁻²

How do weights become wealth? Allocate the fraction w_i of initial wealth W_0 to asset i , buying $n_i = w_i W_0 / S_i(0)$ shares. Holding those shares fixed gives:

$$W_t = \sum_i n_i S_i(t) = W_0 \sum_i w_i \frac{S_i(t)}{S_i(0)}.$$

If two equal allocations experience price factors 1.20 and 0.90, the wealth factor is $0.5(1.20) + 0.5(0.90) = 1.05$. The current wealth fractions change as prices move, even though the share counts stay fixed.

Why do portfolio calculations use a weighted growth rate? Over one short interval, substitute the asset price factors $e^{g_i \Delta t}$ into the wealth ratio:

$$\frac{1}{\Delta t} \ln(W_{\Delta t} / W_0) = \frac{1}{\Delta t} \ln \left(\sum_i w_i e^{g_i \Delta t} \right) \approx \sum_i w_i g_i = g_p.$$

The approximation uses $e^x \approx 1 + x$, $\ln(1 + x) \approx x$, and weights summing to one. It is useful for small interval changes; the wealth formula supplies the exact buy-and-hold calculation. Dirichlet sampling generates candidate long-only weight vectors. Equal concentration parameters of one give uniform sampling over the simplex, while larger equal concentrations favor weights near equal allocation. Comparing sampled portfolios explores possible allocations; optimization in L6a determines the minimum for the specified constraints.

L6a: Data-Driven Minimum-Variance Portfolios

We choose weights using expected growth and growth-rate covariance. The closed-form results allow short positions; the long-only problem uses a target growth floor and is solved numerically.

Symbol	Meaning and context	Units
$\boldsymbol{\mu}_g$	Length- M vector of population mean growth rates.	yr^{-1}
$\hat{\mathbf{g}}$	Estimated mean growth vector, also written $\mathbf{g}'$ in L5b.	yr^{-1}
$\boldsymbol{\Sigma}_g, \hat{\boldsymbol{\Sigma}}_g$	Population and sample growth covariance matrices; $M \times M$.	yr^{-2}
$g_p, \mathbb{E}[g_p]$	Portfolio growth proxy and its mean: $\mathbf{w}^\top \boldsymbol{\mu}_g$.	yr^{-1}
$\sigma_{g,p}$	Growth-rate standard deviation: $\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma}_g \mathbf{w}}$.	yr^{-1}
$g_\star$	Target mean growth; equality in the unconstrained frontier derivation and a lower bound in the long-only problem.	yr^{-1}
$\mathbf{w}_{\text{GMV}}$	Fully invested global minimum-variance allocation.	1
$\mathcal{L}, \lambda$	Lagrangian and equality-constraint multiplier in the GMV derivation.	yr^{-2}
a	Frontier coefficient $\mathbf{1}^\top \boldsymbol{\Sigma}_g^{-1} \mathbf{1}$.	yr^2
b	Frontier coefficient $\mathbf{1}^\top \boldsymbol{\Sigma}_g^{-1} \boldsymbol{\mu}_g$.	yr
c	Frontier coefficient $\boldsymbol{\mu}_g^\top \boldsymbol{\Sigma}_g^{-1} \boldsymbol{\mu}_g$.	1
$d = ac - b^2$	Frontier denominator; positive for the nondegenerate frontier in the lecture.	yr^2
$\sigma_g^2(g_\star)$	Minimum growth variance at target $g_\star$ on the unconstrained frontier.	yr^{-2}
$\mathbf{r}, r_p$	Interval log-return vector $\Delta t \mathbf{g}$ and weighted log-return proxy $\Delta t g_p$.	1
$\boldsymbol{\Sigma}_r$	Interval log-return covariance: $(\Delta t)^2 \boldsymbol{\Sigma}_g$.	1
$\sqrt{\mathbf{w}^\top \mathbf{C} \mathbf{w}}$	Corresponding diffusion-volatility scale, using $\mathbf{C} = \Delta t \boldsymbol{\Sigma}_g$.	$\text{yr}^{-1/2}$

What does the GMV formula solve? We supply the growth covariance $\boldsymbol{\Sigma}_g$ and choose fully invested weights to minimize $\mathbf{w}^\top \boldsymbol{\Sigma}_g \mathbf{w}$. For a positive definite covariance matrix and unrestricted signed weights, the global minimum-variance (GMV) solution is:

$$\mathbf{w}_{\text{GMV}} = \frac{\boldsymbol{\Sigma}_g^{-1} \mathbf{1}}{\mathbf{1}^\top \boldsymbol{\Sigma}_g^{-1} \mathbf{1}}.$$

The numerator adjusts the allocation for variances and covariances; the denominator makes its weights sum to one. For two uncorrelated assets with growth variances 1 and 4yr^{-2} , the weights are $(1, 1/4)/(1 + 1/4) = (0.8, 0.2)$. The lower-variance asset receives more of the budget.

What if we also require mean growth? Add the equality $\mathbf{w}^\top \boldsymbol{\mu}_g = g_\star$. Using the coefficients a, b, c, d defined above, the minimum variance at that target is:

$$\sigma_g^2(g_\star) = \frac{ag_\star^2 - 2bg_\star + c}{d}, \quad d > 0.$$

This curve lets us compare the least variance compatible with different growth targets. The long-only problem instead imposes $w_i \geq 0$ and a growth floor; we solve it numerically. Substituting estimated inputs gives an allocation optimized for those estimates. Its later performance still requires a check on prices reserved for testing.

L6b: Single Index Model and Estimation

The single index model (SIM) relates an asset's growth rate to the market growth rate. Here the subscript M means market; the number of portfolio assets is $|\mathcal{P}|$.

Symbol	Meaning and context	Units
$g_{i,t}, g_{M,t}$	Asset and market growth rates on aligned observation t .	yr^{-1}
N, t	Number of aligned observations and observation index.	Counts
α_i	Intercept of the asset's linear projection on market growth.	yr^{-1}
β_i	Market exposure: $\text{Cov}(g_i, g_M) / \text{Var}(g_M)$.	1
$\varepsilon_{i,t}$	Asset-specific residual growth after removing the intercept and market contribution.	yr^{-1}
ρ_{iM}	Correlation between asset and market growth.	1
$\sigma_{g,i}, \sigma_{g,M}$	Asset and market growth-rate standard deviations.	yr^{-1}
$\sigma_{g,\varepsilon,i}^2$	Population residual growth variance for asset i .	yr^{-2}
$\mathbf{y}, \hat{\mathbf{y}}$	Length- N vectors of observed and fitted asset growth rates.	yr^{-1}
$\hat{\mathbf{X}}$	$N \times 2$ augmented design matrix, with columns of ones and market growth rates.	Columns: 1; yr^{-1}
$\boldsymbol{\theta}_i, \hat{\boldsymbol{\theta}}_i$	Parameter vector $(\alpha_i, \beta_i)^\top$ and its least-squares estimate.	Entries: yr^{-1} ; 1
$\mathbf{r}_i$	Fitted residual series for asset i : $\mathbf{y} - \hat{\mathbf{y}}$.	yr^{-1}
$\text{SE}(\hat{\alpha}_i), \text{SE}(\hat{\beta}_i)$	Standard errors of the fitted coefficients.	yr^{-1} ; 1
R^2	Fraction of sample growth variation explained by the fitted regression with an intercept.	1
g'_i	Sample mean of the observed growth rates for asset i .	yr^{-1}
$g'_M, s_{g,M}^2$	Training-period sample mean and sample variance of market growth.	yr^{-1} ; yr^{-2}
$s_{g,\varepsilon,i}^2$	Residual variance estimate: $\ \mathbf{r}_i\ _2^2 / (N - 2)$.	yr^{-2}

What does an asset's beta tell us? The single index model separates the fitted response to market growth from an asset-specific residual:

$$g_{i,t} = \alpha_i + \beta_i g_{M,t} + \varepsilon_{i,t}, \quad \beta_i = \frac{\text{Cov}(g_i, g_M)}{\text{Var}(g_M)} = \rho_{iM} \frac{\sigma_{g,i}}{\sigma_{g,M}}.$$

Beta includes both correlation and the ratio of growth-rate scales. With correlation 0.60 and asset variability twice the market's, beta is 1.20. A 0.10 yr^{-1} change in market growth then changes fitted asset growth by 0.12 yr^{-1} . The residual describes the remaining variation.

How much variation did the fit explain? Compare the squared residuals with the squared deviations from the asset's sample mean g'_i :

$$R^2 = 1 - \frac{\|\mathbf{r}_i\|_2^2}{\sum_{t=1}^N (g_{i,t} - g'_i)^2}.$$

If the fit leaves 40% of the original sum of squared deviations, $R^2 = 0.60$. This is an in-sample measure for a regression with an intercept. Forecast usefulness must be assessed on later observations, and beta alone does not describe all of the asset's risk.

L6b: SIM Portfolio Inputs and Risk Decomposition

The SIM covariance separates a common market contribution from asset-specific residual variances. Its diagonal residual matrix imposes zero cross-asset residual covariance.

Symbol	Meaning and context	Units
α, β	Intercept and beta vectors, each of length $ \mathcal{P} $.	$\text{yr}^{-1}; 1$
ε	Vector of asset residual growth rates for one observation.	yr^{-1}
$\mu_{g,M}, \sigma_{g,M}^2$	Population mean and variance of market growth.	$\text{yr}^{-1}; \text{yr}^{-2}$
$\mathbf{D}_g$	Diagonal matrix with entries $\sigma_{g,\varepsilon,i}^2$.	yr^{-2}
Ω_ε	General residual covariance matrix, allowing nonzero off-diagonal entries.	yr^{-2}
$\hat{\mathbf{D}}_g$	Diagonal matrix of fitted residual variances $s_{g,\varepsilon,i}^2$.	yr^{-2}
$\hat{\mathbf{g}}_{\text{SIM}}$	Fitted mean vector: $\hat{\alpha} + \hat{\beta}g'_M$.	yr^{-1}
$\hat{\Sigma}_{g,\text{SIM}}$	Fitted SIM growth covariance, using the same training observations as the coefficient fits.	yr^{-2}
β_p	Portfolio market exposure: $\mathbf{w}^\top \beta$.	1
ε_p	Portfolio residual: $\mathbf{w}^\top \varepsilon$.	yr^{-1}
$\bar{\sigma}^2$ (residual bound)	Common upper bound on residual variances in the diversification argument.	yr^{-2}

How does the market model supply portfolio inputs? Assume residuals have zero mean, are uncorrelated with market growth, and are uncorrelated across assets. Taking means and covariances of the vector SIM gives:

$$\begin{aligned}
 \mu_g &= \alpha + \beta \mu_{g,M}, \\
 \Sigma_g &= \underbrace{\sigma_{g,M}^2 \beta \beta^\top}_{\text{shared market contribution}} + \underbrace{\mathbf{D}_g}_{\text{residual variances}}, \\
 \hat{\Sigma}_{g,\text{SIM}} &= s_{g,M}^2 \hat{\beta} \hat{\beta}^\top + \hat{\mathbf{D}}_g.
 \end{aligned}$$

The last line substitutes training estimates for population quantities. Off-diagonal covariance comes from the shared market input; the diagonal also includes each asset's residual variance. These estimates can replace the empirical inputs in the L6a allocation problem.

Which risk can diversification reduce? For fixed weights, multiply the covariance by the weights on both sides:

$$\text{Var}(g_p) = \underbrace{\beta_p^2 \sigma_{g,M}^2}_{\text{market risk}} + \underbrace{\sum_i w_i^2 \sigma_{g,\varepsilon,i}^2}_{\text{residual risk}}.$$

If $|\mathcal{P}|$ equally weighted assets have the same residual variance v , residual risk is $|\mathcal{P}|(1/|\mathcal{P}|)^2 v = v/|\mathcal{P}|$. If their betas also coincide, their common market exposure remains. Nonzero residual covariances add cross terms omitted by the diagonal model, which is why the lecture examines residual dependence. All variances here are growth-rate variances in yr^{-2} .

L6b: Risk-Free Allocation and the Capital Allocation Line

We combine a risky portfolio with an asset whose growth rate is known over the holding period. The risk-free weight controls lending and borrowing.

Symbol	Meaning and context	Units
g_f	Known growth rate of the risk-free asset over the matched holding period.	yr^{-1}
w_f	Risk-free wealth fraction; negative values represent borrowing.	1
p, c (subscripts)	Risky portfolio and complete portfolio that combines risky and risk-free positions.	Portfolio labels
g_p, g_c	Growth-rate proxies for the risky and complete portfolios.	yr^{-1}
$\sigma_{g,p}, \sigma_{g,c}$	Corresponding growth-rate standard deviations.	yr^{-1}
SR_p	CAL slope: $(\mathbb{E}[g_p] - g_f)/\sigma_{g,p}$ for the observation interval.	1
$\tau = 1 \text{ yr}$	Reference year used when annualizing the Sharpe ratio.	yr
$\text{SR}_p^{\text{annual}}$	Annualized ratio: $\sqrt{\tau/\Delta t} \text{SR}_p$ under independent-increment scaling.	1
$\mathbf{e}$	Excess-growth vector: $\boldsymbol{\mu}_g - g_f \mathbf{1}$.	yr^{-1}
$\mathcal{T}, \mathbf{w}_{\mathcal{T}}$	Tangent risky portfolio and its weights, normalized to sum to one.	Label; 1
$\mathbf{w}$ (complete allocation)	Risky weights as fractions of total initial wealth; $\mathbf{1}^\top \mathbf{w} = 1 - w_f$.	1
$g_\star$	Required mean growth of the complete portfolio.	yr^{-1}

What happens when we lend or borrow? Put fraction w_f in the risk-free asset and fraction $1 - w_f$ in a fixed risky portfolio. Its known growth contributes to the mean but contributes no variance. Thus:

$$\mathbb{E}[g_c] = g_f + (1 - w_f)(\mathbb{E}[g_p] - g_f), \quad \sigma_{g,c} = |1 - w_f| \sigma_{g,p}.$$

For daily observations ($\Delta t = 1/252 \text{ yr}$), suppose $g_f = 0.02 \text{ yr}^{-1}$, $\mathbb{E}[g_p] = 0.08 \text{ yr}^{-1}$, and $\sigma_{g,p} = 3.0 \text{ yr}^{-1}$. As in L3a, this standard deviation describes daily growth rates expressed per year. A 50% risk-free allocation gives mean growth 0.05 and standard deviation 1.5 yr^{-1} . Borrowing 50% of initial wealth gives $w_f = -0.5$: mean growth becomes 0.11 and standard deviation 4.5 yr^{-1} . Borrowing increases both risky exposure and uncertainty.

Which risky portfolio defines the best capital allocation line? The tangent portfolio maximizes excess mean growth per unit standard deviation. With positive definite $\boldsymbol{\Sigma}_g$, signed risky weights allowed, and GMV mean growth exceeding g_f , the lecture obtains:

$$\mathbf{w}_{\mathcal{T}} = \frac{\boldsymbol{\Sigma}_g^{-1} \mathbf{e}}{\mathbf{1}^\top \boldsymbol{\Sigma}_g^{-1} \mathbf{e}}, \quad \mathbf{e} = \boldsymbol{\mu}_g - g_f \mathbf{1}.$$

This selects the risky mix; the risk-free weight then sets total exposure. The borrowing example assumes the same known rate for borrowing and lending. Long-only bounds, position limits, or borrowing limits change the feasible choices and are included in the numerical allocation problem.

Symbols Whose Meaning Changes with Context

Use the lecture and any subscript together with the symbol. This index collects common collisions; the preceding tables supply definitions and units.

Symbol	Contexts to distinguish	Where
α	Tail index; regression significance level; Dirichlet concentration; SIM intercept.	L3a; L4b; L5b; L6b
δ	Option sensitivity in the orientation preview; growth innovation δ_k in EMA.	L1a; L5a
λ	EMA decay factor; GMV equality-constraint multiplier.	L5a; L6a
μ, μ_g	Drift of expected share price; mean log growth. Under GBM, $\mu_g = \mu - \sigma^2/2$.	L4b-L5b
σ_g, σ	Standard deviation of observed growth rates; diffusion volatility. They satisfy $\sigma = \sqrt{\Delta t} \sigma_g$ under the matched GBM convention.	L3a-L6b
$C, \mathbf{C}$	Cash coupon; covariance-rate matrix. Boldface and context distinguish them.	L2a; L5b
$\Sigma_g, \mathbf{C}$	Growth covariance; covariance rate, with $\mathbf{C} = \Delta t \Sigma_g$.	L5b-L6b
$D, D(0, t), D_{\text{mod}}, D_{\text{mac}}, \mathbf{D}_g$	Days to bill maturity; discount factor; duration measures; observed down-multiplier set; diagonal residual covariance.	L2a; L2b; L3b; L6b
n	Compounding periods per year; total tail-analysis sample size.	L1b-L2a; L3a
d	Bill discount quote; lattice down factor; frontier coefficient $ac - b^2$.	L2a; L3b; L6a
K, K_N	Bond convexity; terminal price target; binomial up-count K_N .	L2b; L5a; L4a
M	Growth-magnitude bound; Monte Carlo path count; asset count; market subscript.	L3b; L4b; L5b; L6b
T, t	L3a uses T for a count of prices; lattice t is a step count; continuous-time models use time in years.	L3a-L6b
p	Up probability; number of regression parameters; risky-portfolio label.	L3b; L4b; L6b
$\rho(k), \rho_T, \rho_{ij}$	Autocorrelation; scaled NPV (including L4a node function $\rho(k)$); cross-asset correlation.	L3a; L4a; L5b
$W(t), W_t$	Wiener process; portfolio wealth.	L4b; L5b
$\mathbf{r}$	Regression residuals in L4b and L6b; interval log-return vector in L6a. Units follow the response variable.	L4b; L6a; L6b
$\hat{\mathbf{X}}$	Design matrix with time as predictor; design matrix with market growth as predictor.	L4b; L6b
τ	Lattice lag; real-valued up-count threshold; reference year for annualizing a Sharpe ratio.	L3b; L4a; L6b
$\nu, \nu_{j,k}$	Student- t degrees of freedom; regression residual degrees of freedom; cash-flow direction.	L3a; L4b; L1b
$X, X_j, \mathbf{X}_t, \mathbf{X}$	Growth magnitude; bond cash flow or lattice up indicator; branch-count vector; worked-example noise vector.	L3a; L2b/L3b; L4a; L5b
$c, c_{j,k}$	Coupon rate; frontier coefficient; complete-portfolio subscript; cash-flow amount.	L2a-L2b; L6a; L6b; L1b

Keeping the reference current. Each week, we will add new symbols with their definitions, units, and lecture context. Please flag a missing or unclear entry with the lecture label and the equation where it appears.